Bangladesh University of Engineering and Technology
Department of Computer Science and Engineering

Course: CSE 206
Digital Logic Design Sessional

Experiment No. 9
Flip-Flops

Design and implement the following problems:

1. Design a 3-bit **synchronous** counter using **D flip-flops** and basic gates that counts in the following order:
   0 → 1 → 3 → 7 → 6 → 4 → 0
   You can use IC 7474 (D flip-flop) for this problem.

2. Design a **2-bit asynchronous mod-4 down counter** using **T flip-flops**. You can use IC 7476 (JK flip-flop) as T flip-flop for this problem (using J=K).

Lab Sheet: For each of the problems, your lab sheet should cover
- Problem definition
- State diagram and state transition table (for synchronous counter)
- Simplified boolean expressions
- Circuit diagram with pin number
- Required instruments for implementation

Answer the following questions:
1. Design a 3-bit synchronous Gray code counter using JK flip-flops. (You must not convert JK to T)
2. Design a synchronous BCD counter using D flip-flops.

Please elaborate all steps including state diagram and state transition table.